Industrial Flare Stack MOD for CSS Satisfactory



"*Industrial Flaring, just like the real plant*"

In 1978 I started with programming on a TI-99 home computer as they called it in those days, using Assembler and Basic language. The goal was to control a satellite receiving unit to acquire images with, for those days, incredible resolution. Even back then, you could make out a chair in a garden if you knew which satellite to track and how to follow it from horizon to horizon. In 1983 it changed to a C64, and I continued programming what we called demos in those days: just showing off the graphics capability of an 8-bit computer that was incredible at that time. Going from a 286 to a 386, a Pentium, an Intel i7/GTX 1080, and now to an AMD 9950X3D/RX 7900 XTX.

I was raised on Wolfenstein 3D / Mein Leben, Doom, Duke Nukem 3D, Quake, Unreal (the game), and Half-Life. After that era, a lot of new games landed, but I lost interest because they weren't truly new. Then in 2019 Satisfactory came along. This was a genuinely new type of game: 3D first person factory automation, and up to today, I think it is just a wonderful game, both in terms of gameplay and the immersion you feel while playing and the incredible mod ability.

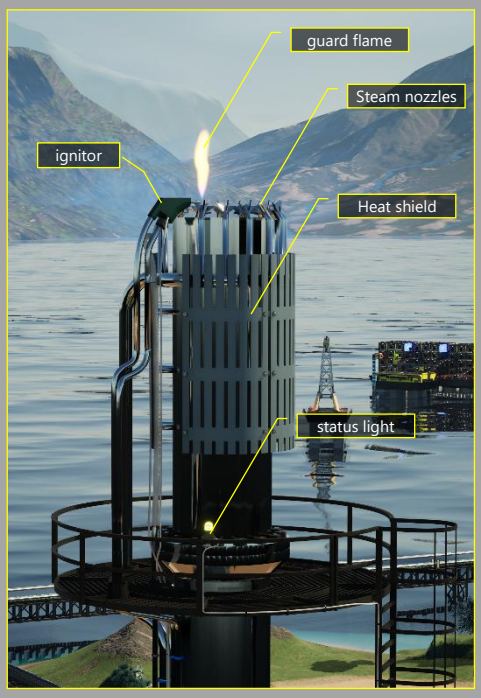
In October last year I started considering dusting off my programming skills to create a MOD for this game as it was extremely appealing to my creative drive…. I wanted to create a serious industrial Flare Stack to flare off combustibles. For Satisfactory it can support the event of a downstream blockage, to keep energy production running by all means, but also the immersion to see and hear such thing is just awesome (well that's what I think).
Unreal Engine with its Blueprint coding was far from what I thought was real programming… but then I found out that actually this BP coding combined with C++ gives you 100% freedom in Unreal Engine to do whatever you want. I struggled lots of times to get a handle on it. One of the worst parts was to port a CAD design to a game. Professionally I use a lot of parametric design software like AutoCAD Inventor. But the result for a game is a disaster: millions of triangles!

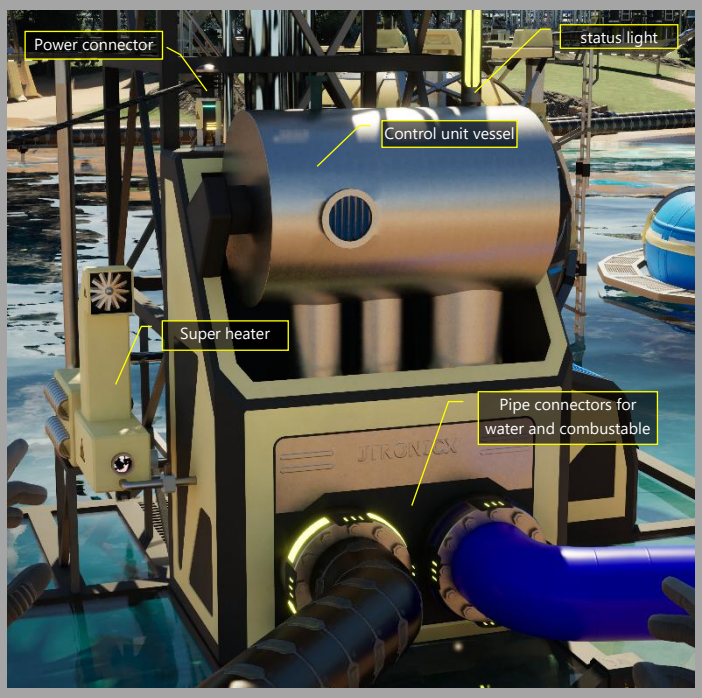
So then came Blender. OMG, that was the biggest inconvenience I have ever experienced, but when you get a grip on it, it's just awesome.

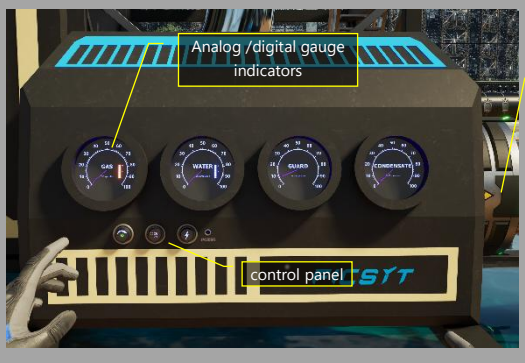
Using all these tools I created this MOD for Satisfactory and I put a lot of time and effort in it to give you a "close-to-real" experience to control off-spec chemical products or get rid of oversupply of feed stock.
On an oil rig there is a lot of shale gas coming out of the oil well. This has to be flared off: you cannot just emit pure methane into the atmosphere. This is dangerous (the whole rig can explode) and it's an environmental crime. Same is true with chemical plants like crackers where they crack ethane or Naphtha to make ethylene and propylene for e.g. plastics and higher molecular weight products like fuels and thousands of other intermediates. If the product goes off-spec you need to burn it in a safe way, else the whole factory blows up.
Flaring is one of the crucial processes in the energy and chemical industry to assure a safe and continuous production output so I think FICSIT should be thrilled to see it in action!

Flare Stack component overview









Component description

Control unit It has the power connector (daisy-chain capable), the machine status light and 2 fluid pipe flanges to connect water and a combustible fuel (see page 5 for recipes). The Flare Stack identifies on which flange the fuel is connected and this flange will light up yellow (when connected to power). There is also a “liquid seal drum / knock down drum” (vessel) which in real life does the vapor/liquid separation and prevents flashbacks of flames, this is not a real-life implementation, which is far more complicated and in fact consists out of 2 separate vessels. When active, you can see though the glass window some action.

Super heater Produces the required steam for the 14 nozzle steam injector on the Flare head. It uses an advanced fuel-to-plasma conversion process to generate the necessary thermal energy for steam production. The burner has a glass window and air intake ventilator for visual confirmation of its operation. The super heater will self-regulate the required steam and switches off when non water recipes are in use.

Condensate Unit Acts as the primary waste management and monitoring unit for your Flare Stack system. It handles the byproduct liquids dropped out during gas processing (condensate), safeguards the operation from overflowing, and houses the primary user interface.
An internal reservoir catches up to 100 cubic meters of condensate.
At 100% capacity, it triggers a stop of the Flare Stack operation. An external Drain Flange connection enables piping the waste fluid into larger industrial storage tanks, allowing you to scale up your storage capacity indefinitely. Future extension will allow you to process this waste into usable materials. On its front are 4 gauge scales that show the machine status: combustion rate and buffer amount of the combustible (gas) & water, a guard flame and waste buffer gauge for visual confirmation that it is active. a fully interactive control panel consisting of 3 turn/push knobs and a low fuel indicator (usage explained on page …)

Stairway Can be accessed from the back of the super heater. There are 16 in total up to the lower deck. At every second turn you will find a machine status light in the frame corner that gives you the actual operation mode status of the Flare Stack.

Lower deck at 80 m you arrive on the lower deck. This is a great viewing location but a bit windy. A small cage ladder gives access to the upper deck.

Upper deck A circular deck at 85 m, here it's rather windy and extremely noisy if the main Flare is active. You will see the various pipes for steam, guard gas, condensate and fireball (manual ignition) and the ignition head. Audio sounds will warn you of a too close proximity to the ignitor head when you "fly up". Be careful of the heat shield as this will induce thermal burn and health damage. Do not touch the ignitor head when the Flare Stack is on power and the guard flame is burning. You will die instantly…

Ignitor Located on the rim of the main Flare mouth is a ceramic ignitor head. 2 electrodes will create a spark for the auto ignition process. If you want to see it, you need to fly up close to the head and then attach the power cable from that distance.



Steam nozzles Around the rim of the Flare mouth are 14 steam nozzles, they operate when you run a recipe using water and inject high pressure steam into the Flare tip. This increases the burn efficiency and thus reduces the smoke amount during flaring. The Flare Stack detects when you run out of water and automatically switches to the non-water variant for the same fuel type. Visually you will notice this also from a very far distance as the smoke amount will be drastically increased.

Operating the Flare Stack





Unlocking: in the HUB Tier 8 you will find a new section called Industrial process. After unlocking the Flare Stack, the Flare Stack and the Telemetry Junction can be accessed from the build menu in the logistics section under 8 Process Control.

The Flare Stack behaves like any other Satisfactory machine except the behavior of loading of fluids. Initial build is in standby mode and the internal buffers will fill up if fluid pipes are connected. The production state will be saved at level save and is retained a loading. The combustibles that can be used with their burn speed, waste production and water requirements are listed in the table below. The "dirty" recipes are the variants without water.

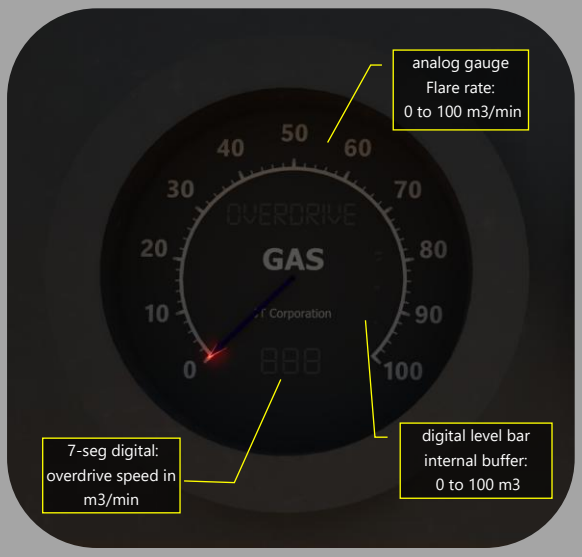
Table 1 Flaring consumption and condensate output (values at 100% production speed)

Recipe	Combustible[1)] (m³/min)	Water (m³/min)	Condensate (m³/min)	Flare Smoke[2)]
Oil	100.0	50.0	50.0	1.0
dirty Oil	100.0	0.0	25.0	15.0
Heavy Oil Residue	100.0	40.0	40.0	1.5
dirty Heavy Oil Residue	100.0	0.0	20.0	30.0
Fuel	100.0	20.0	20.0	1.0
dirty Fuel	100.0	0.0	10.0	10.0
Liquid Bio Fuel	100.0	15.0	10.0	1.0
dirty Liquid Bio Fuel	100.0	0.0	10.0	10.0
Turbo Fuel	100.0	15.0	10.0	1.0
dirty Turbo Fuel	100.0	0.0	10.0	10.0
Rocket Fuel	100.0	10.0	5.0	0.5
dirty Rocket Fuel	100.0	0.0	5.0	4.0
Ionized Fuel	100.0	5.0	2.5	0.0
dirty Ionized Fuel	100.0	0.0	2.5	1.5

[1)] The guard flame (ignitor of main flare) when its on always consumes fuel at a rate of 0.3 m3 /min. it will shutoff when the amount left in the buffer is 0.9 m3.

[2)] The values are an indication of the amount of smoke produced by the main Flare compared to the oil recipe.

To activate the Flare Stack use the standard machine UI of Satisfactory or the interactive control panel on the front of the condensate unit.



A gauge scale and its indicators





Point mouse pointer



Press RMB while pointing

When in near proximity of the panel, pointing the mouse pointer at a knob the base ring with light up soft orange. When press and holding the RMB that knob will become active (glowing red base ring). While holding RMB you can then use LMB and MW to select and set the functions.

The Mode / Config knob

Default mode shows the game time and local time on 2 rows of 5x7-seg displays. All data is displayed as on real 7-seg displays so text is displayed in 7-seg characters.

MW up/down scroll through menu's:

Main:	Mode – PID – Config
Mode (clock mode):	24hr-12hr- back
	Select mode with MW and confirm
PID (Auto tune):	Auto – Man – <status>- back
	Here you can start auto tuning of Flare Stack or manual mode (see next page)
Config (parameters):	Smoke – PID SL – PID Kp – LOG – UI COL - back
Smoke:	set Smoke dwell time between 6 and 30 sec
PID SL:	set PID SlopeGain value(see next page)
PID Kp:	set PID PressureKp value (see next page)
LOG:	set LOG file FALSE or TRUE (see next page)
UI COL:	UI Red – UI Grn – UI Blue – UI Int
	Select a color or intensity and use MW to change value, will update in real time, press to confirm

Auto tune the Flare Stack

The Flare Stack has a dynamic flow monitoring option for automatic tuning its burn rate in combination with the Telemetry Junction.
To use it effectively use the following procedure:

1. If not yet built, use the Telemetry junction instead of the standard junction to drag off the fuel into the Flare Stack. This junction will monitor the flow, change in flow and pressure between the upstream and downstream connection.
2. First make sure your production plant that is using the fuel source you want to control is in a stable production state / the state you want to monitor as reference situation (there are various options).
3. Switch on the Flare Stack and set its burn rate to ~0 or at a certain consumption rate to buffer for planned downstream expansion.
4. Now select the PID menu and then select AUTO and press the knob. The Display should now show “Gating” and all three ring indicators on the Telemetry Junction flash red. After ca. 10 seconds the ring colors switch to solid Green (Upstream), Red (Downstream) and Yellow (Flare Stack) and the display shows “stabilize”. The red Antenna indicator on the Junction will flash at low rate.
   During ca. 60 seconds the PID control will measure the actual flow behavior over the Junction. It will register the overall slosh profile in the pipe and the real average of all down stream consumers.

   When it is ready it will switch to “active” and the red Antenna indicator will pulse at high rate. and your Flare Stack will slowly start to respond depending on the reference state. During the first few minutes the amplitude of regulation can intensify but it will find a balance that represents the over or under consumption. The Flare Stack will now regulate so any change on the consumer demand is compensated. One Flare Stack can regulate a variation of up to 250 m3/min

The values of SL and Kp are default on 2.0 and 5.0.
They are arbitrary found based on a rather complex downstream setup that had also upstream producers on the downstream consumer line.

Auto Tuning and fluids in Satisfactory

Coffee Stain Studios made a smart implementation for fluids; it is impossible to make an actual continuous flow calculation in a game with 100.000 or more machines without wrecking even an AMD 9950X3D.

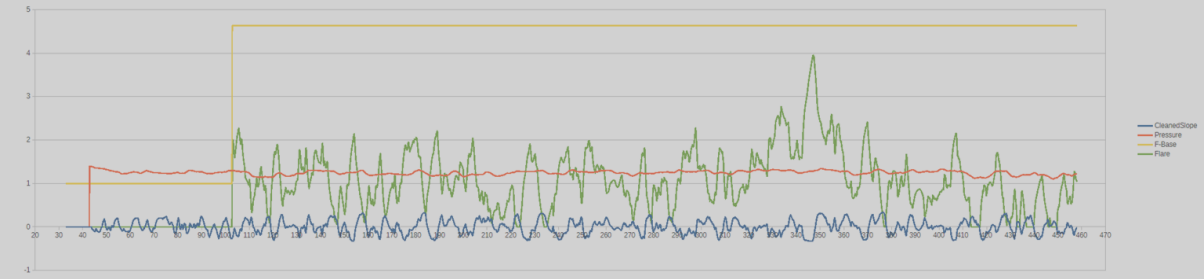
So Fluids are not actually consumed continuously like you may think. Each producer or consumer instantly pushes / pulls a certain amount of fluids/gas in/out the machine, and that at every start of its production cycle. This is why we see the ”slosh” effect. When you measure flow in Satisfactory pipes, you will see all those peaks of these pulls/pushes. My Algorithm catches this slosh profile and can then calculate the real effective flow in a pipe. But in order to keep the Flare Stack somewhat reactive to change in this actual flow, It regulates always around the target point with a certain under and over shoot, but in such way it regulates fast to the target average but still can see a change in downstream demand (~ 15 sec) so it adapts its burn rate before a up stream producer throttles down…. This was the main objective for this auto tuning.

If you like to experiment with the SlopeGain and the Pressure Kp of this algorithm, there is a LOG function under the config menu where you can activate the creation of a log file during the auto tuning process.
This log file is saved in your save folder under the Log folder.

File name is: FSPIDLog_yyyymmdd_hhmmss.csv

In the graph below an example of the 5 parameters you can study.
F-Base: this is the avg consumption found after the calibration period
Pressure: fill ratio in Junction.
Cleaned Slope : change in flow
Flare: Flare Stack control speed

In this example the steady flow was 4.63 m3/sec or 278 m3/min. but on consumer was off: ~ 80 m3/min so the oil well was under-producing: thus the Flare Stack started to consume: about 1.24 m3/sec. (~75 m3/min) to compensate and keep the oil well running at 100% without starving downstream.



Simplified algorithm:

FlareTargetSec = (SteadyBurn + PressureBurst) * PressureScale
SteadyBurn = (F_base - F30) - (SlopeGain * S30)
PressureBurst = (P10 - 1.0) * PressureKp

PressureScale: an initial attenuation of amplitude during fire up.

If you have any suggestion, question do not hesitate to write me a mail or catch me up on discord, I appreciate any feedback if you like it!

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Troubleshooting

Troubleshooting Auto-Tune "Gating" Failures:

If the Telemetry Junction fails to transition from "Gating" to "Stabilize" or if the calibration sequence aborts, it is typically caused by extreme fluid sloshing or severe flow instability in the pipe network during the 60-second measurement window. To fix this, let your upstream production line run continuously for a few minutes to allow pressures and buffer levels to settle into a steady rhythm before initiating auto-tune, and ensure your pipeline is not bottlenecking or slamming violently against flow limits. False Gating will normally not happen but under extreme conditions as described above it could happen: then put auto mode off (set to man) and restart (put back on auto mode)

Power outage or a blown fuse:

The Flare Stack's internal buffers will hold steady, but active flaring and the super heater will pause until power is successfully restored. Finally, when running continuous high-waste volume recipes such as Oil or Heavy Oil Residue, it is recommended to chain multiple industrial fluid buffers to the external Drain Flange to safely scale your waste capacity until I deliver the thermal cracker. You CAN flush the storage but …… well …… neeehhhhhhh !